

MATH0074 Topology and Groups

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0074
<i>Level:</i>	7 (UG/PG)
<i>Normal student group(s):</i>	UG Year 3 or 4 Mathematics degrees PG: MSc Mathematics
<i>Value:</i>	15 credits (= 7.5 ECTS credits)
<i>Term:</i>	1
<i>Assessment:</i>	90% examination and 10% coursework
<i>Normal Pre-requisites:</i>	MATH0051, MATH0053
<i>Lecturer:</i>	Dr Y Naqvi

Course Description and Objectives

Topological spaces are one of the fundamental objects studied in pure mathematics. The first part of this course is a more detailed investigation of topological spaces (first introduced in Analysis 4), including the important construction of quotient spaces, which allows us to build new spaces by gluing old ones together.

We will then begin the study of ‘topological invariants’, which are tools used to tell topological spaces apart. Studying paths will lead us to connectedness and π_0 , our first invariant. Studying loops will lead us to π_1 , the fundamental group, which is a much richer and more interesting invariant. We will learn how to compute π_1 in many examples, and see the deep connection between the fundamental group and covering spaces.

Recommended Texts

- (i) J. Munkres, *Topology*, 2nd Edition (Pearson, 2017), Chapters 1–3, 9–14.
- (ii) William S. Massey, *A Basic Course in Algebraic Topology* (Springer-Verlag, 1991), Chapters 2–6.
- (iii) John Stillwell, *Classical Topology and Combinatorial Group Theory* (Springer-Verlag, 1993).
- (iv) A. Hatcher, *Algebraic Topology* (Cambridge University Press, 2001), Chapter 1.

Detailed Syllabus

- Review of topological spaces, continuous maps, the Hausdorff condition, compactness.
- Product spaces and disjoint union.
- Quotient spaces. Compactness of products and quotients.
- Review of connectedness and path-connectedness.
- Connected components. π_0 as a topological invariant.
- Homotopies, homotopy equivalence, contractibility. Invariance of π_0 .

- Fundamental groups. Fundamental group of S^1 .
- Behaviour under continuous maps. π_1 as a homotopy invariant.
- Covering spaces, path lifting and homotopy lifting, covering transformations.
- Covers and subgroups. The universal cover.
- π_1 of a wedge product of circles. Other examples of Seifert-Van Kampen theorem.